Biological and climatic constraints drive reproductive synchrony

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Abstract

For many tree species, successful regeneration is thought to depend on synchronous reproduction with 'boom-and-bust' cycles—called masting. Climate has been identified as a potential driver of masting timing, raising concern that anthropogenic climate change could disrupt synchrony within forests. Yet how trees synchronize is hard to explain, partly because it requires data and models that scale from individual trees to populations. Here, using long-term seed production data from beech trees across England, we integrate climatic cues with a reproductive constraint within each tree: because floral bud initiation and fruit maturation overlap in time, investment in a heavy crop comes at the expense of the next. Our results support the idea of masting as a synchronous 'boom-and-bust' cycle, with 87% of trees within each population sharing the same reproductive state in any given year since 1980. Synchrony between populations did not decline significantly over the last decade despite marked warming (84% before 2015 versus 83% after). Warm summers increased the probability that trees transitioned into a high reproductive state, but individual-level reproductive constraints prevented runaway increases in masting frequency, even under forecasts of extreme warming. We found synchrony was driven mainly by cold summers, which pushed most trees into a shared low reproductive state, providing the conditions for synchronous reproduction when a warm summer followed. Our results suggest the current interplay of individual-level constraints and population-level climatic cues will lead to continued synchrony with warming in the immediate future, and highlight the need for more tree-level data and mechanistic models for accurate forecasts.

Introduction

The acceleration of climate change may cause abrupt ecological effects worldwide (Trisos *et al.*, 2020). Increasing extreme conditions could disrupt key ecological processes, and drive ecosystems toward critical transitions (Wernberg *et al.*, 2016). Across biomes, recent work has suggested major shifts in pests and pathogens, while the local extirpation of rare species and increase in invaders across taxa has driven biotic homogenization (Carlson *et al.*, 2025; Dornelas *et al.*, 2014; Daru *et al.*, 2021). Forests in particular appear to be shifting, with lagging elevational and latitudinal range expansions, alongside increases in adult tree mortality (McDowell *et al.*, 2020). This has driven growing concerns about the future of forests to act as carbon sinks globally (Albrich *et al.*, 2020; Forzieri *et al.*, 2022; McDowell *et al.*, 2020), and thus growing pressure to understand the mechanisms that underlie forest regeneration (Foest *et al.*, 2024; Hacket-Pain *et al.*, 2025; Chakraborty *et al.*, 2024; Foest *et al.*, 2026).

Regeneration in many temperate and tropical forests depends on the widely observed phenomenon of masting. Masting—defined by 'boom and bust' cycles of reproduction that are synchronous across most individuals of a population (Janzen, 1978)—occurs across many trees species and is hypothesized to drive fitness benefits during reproduction and the early stages of establishment (Kelly, 1994; Bogdziewicz *et al.*, 2024). The most commonly invoked fitness benefit, 'predator satiation,' hypothesizes that years of high seed production overwhelm seed predators and thus allows a higher proportion of seeds and seedlings to escape predation and establish (Janzen, 1971; Kelly, 1994). Fitness benefits of masting may also operate earlier in reproduction, by increasing pollen exchange and genetic outcrossing (Carlson *et al.*, 2014; Bontrager & Angert, 2019). While these different hypotheses vary in their exact mechanism and life stage, they all imply that the benefits of masting depend on an economy of scale: most individual trees within a population (or location) must be synchronous.

Despite the importance of synchrony to yielding fitness benefits, how synchronous masting is and how it occurs remain poorly understood. Most hypotheses for how synchrony emerges focus on shared environmental cues at the species-level (Kelly & Sork, 2002; Bogdziewicz *et al.*, 2024). Among many potential cues, an abundance of work suggests that ideal climatic conditions during bud formation may lead to masting (Nussbaumer *et al.*, 2018; Journé *et al.*, 2025). In temperate systems, ideal conditions include sufficiently warm temperatures during the growing season before masting—when reproductive buds generally develop—alongside mild spring conditions that do not cause tissue loss after buds form (i.e., no late spring frosts, Vacchiano *et al.*, 2017; Gallego Zamorano *et al.*, 2016). Climate change, however, is rapidly shifting temperatures across the spring and summer, including the prevalence of spring frosts (Zohner *et al.*, 2020; Chamberlain *et al.*, 2021), and thus could disrupt masting dynamics. Such changes would have major implications for forest resilience (Bogdziewicz *et al.*, 2023; Foest *et al.*, 2024), but forecasting them requires better understanding how these climatic cues affect reproductive biology, from buds to fruits.

Most temperate woody species develop reproductive buds through a multi-year process. In the common two-year reproductive cycle (Fig. 1A-B), bud formation begins the year before flowering and seedset—potentially explaining why summer temperatures from the previous year appear to trigger masting (Geber *et al.*, 1997; Wilkie *et al.*, 2008). The following year some of these buds may be lost to spring frost—potentially reducing the number of seeds in what otherwise would have been a masting year. This two year cycle, however, also introduces a constraint in most species (Fig. 1C). Because floral buds for the following year develop simultaneously with current-year fruit (Geber *et al.*, 1997), a large fruit crop can depress bud initiation (through hormonal inhibition and within-plant resource competition, Monselise & Goldschmidt, 1982; Bangerth, 2009; Milyaev *et al.*, 2022). The result is physiological constraints on flower and fruit development that is often invoked to explain why many woody species show alternate bearing—with a large crop production year ('on-year') often followed by one or several years of low production ('off-years', Monselise & Goldschmidt, 1982; Wilkie *et al.*, 2008).

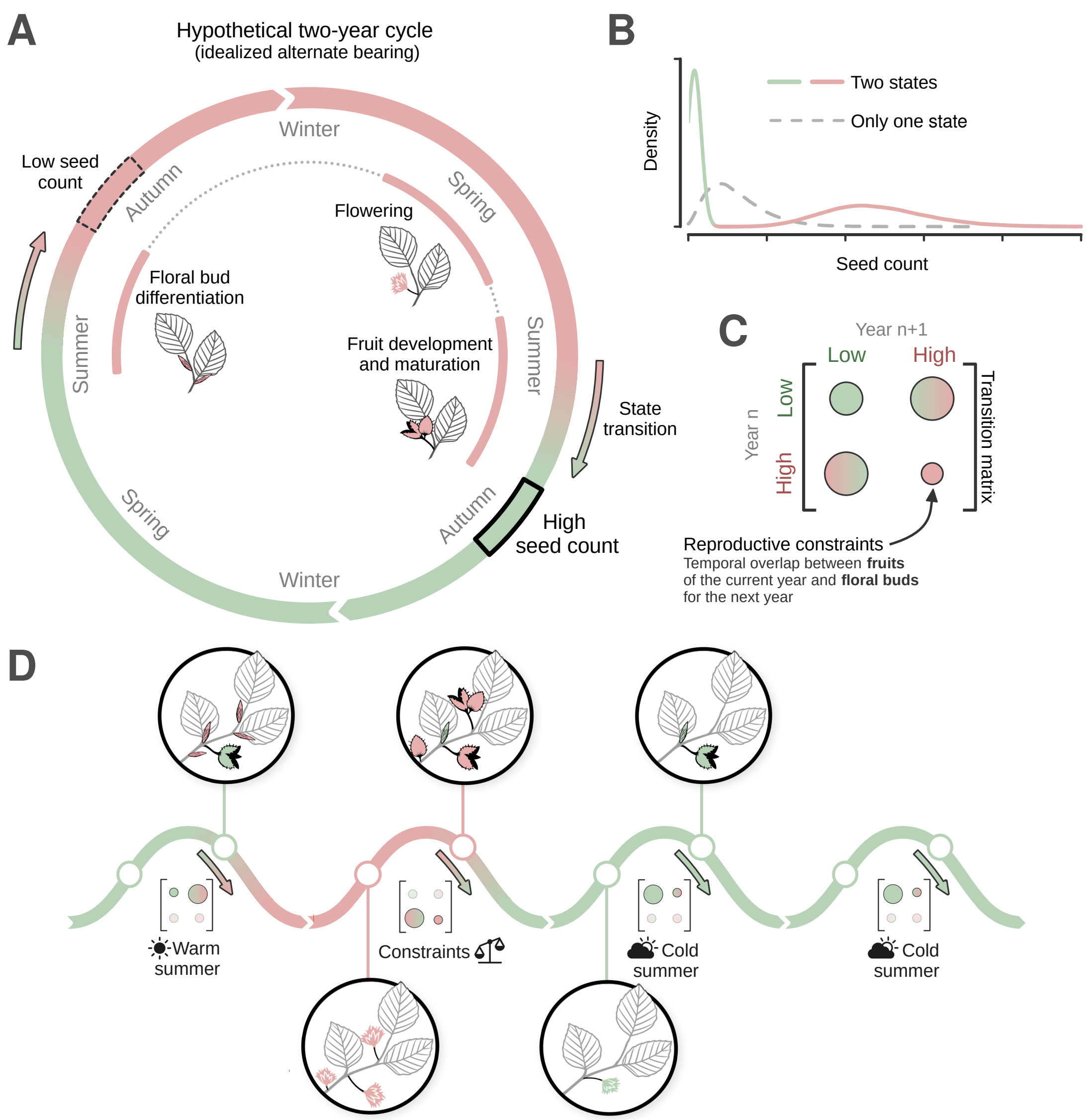

Figure 1: **A two-state model captures the interplay of reproductive constraints and climatic cues. (A)** A large investment in floral buds in summer (initiation and differentiation) leads to a large flowering the following spring. Under favorable conditions, many fruits then develop and mature, producing a high seed count in autumn. Investment in the fruits reduces the investment in floral buds, leading to a low seed count the year after. Green and red represent the low and high reproductive states, respectively, and the arrows represent the transitions between the states. **(B)** The transition between low and high reproductive states produces a bimodal distribution of seed counts, which a single-state model would not explain. **(C)** Transitions between low and high reproductive states are governed by a matrix of transition probabilities. The probability of staying in a high state is likely low, reflecting the reproductive constraint: the temporal overlap between maturing fruit in the current year and floral bud formation for the next year. **(D)** On top of this constraint, climate, and summer temperature in particular, influences the transition probabilities. A warm summer can push a tree into a high reproductive state, but the tree is unlikely to stay in a high state the next year because of the reproductive constraint.

These constraints may affect how climate—and climate change—impacts seed production. Ideal climatic conditions may trigger higher investment in floral buds only for individual trees not already in a high reproductive state (Fig. 1D).Warm summers would therefore not affect all trees equally—trees in a high reproductive state would be unaffected, while those in a low reproductive state would likely increase their probability of transitioning to a high reproductive state. Once a reproductive state has been established, then spring frost may have an effect, with damage from frosts most likely to reduce potential seed production for those trees in a high reproductive state. These individual-level reproductive dynamics may then scale up to generate population-level synchrony, with masting emerging through a combination of climatic variability and reproductive constraints.

Here we investigate how climate affects masting when explicitly modeling how climatic cues interact with individual reproductive constraints to generate population-level synchrony. Using seed data collected on individual trees across populations of beech forests of England since 1980, we first test whether individual trees show evidence for alternative reproductive states that could lead to masting at the population level. Using these states, we then examine how synchrony arises at at the individual level and scales up to populations. Finally, we leverage these results to forecast how future warming may affect both individual tree-level seed production and population-level synchrony.

Results and discussion

Following the paradigm that masting is defined by of 'boom and bust' reproduction, our model identified high and low reproductive years as alternative latent states (Fig. 2). In high ('boom') reproductive years, individual trees produced 168 seeds on average (90% uncertainty range: 163 – 173), while in low ('bust') reproductive years they produced 52 seeds on average (48 – 56 seeds per tree). Seed number per tree in low and high reproductive years, however, overlapped (Fig. 2A), with the difference between these two states thus occurring mostly in the opposite extremes: years in the high reproduction latent state included very high seed counts, which would almost never be possible in the low reproduction state, while years in the low reproduction state were characterized by extremely low counts, including a high probability of no observed seeds at all (Fig. 2A).

Transitioning between reproductive states (high or low) depended on a tree's state in the previous year (Fig. 2B). In a year with average climatic conditions (mean summer temperatures of 19.7℃), a tree in a low reproductive state was most likely to transition to a high reproductive state the next year, with a probability of 64% (61 – 68%), and a 1.8 times lower probability (36%, 32 – 39%) of staying in a low reproductive state (Fig. 2B). Under the same average climate, a tree in a high reproductive state was almost certain to transition to a low reproductive state, with a probability of 99% (97 – 100%), and very unlikely to remain in the same state (1%, 0 – 3%). Trees thus commonly shift into the opposite reproductive state each year, but the low reproductive state is much more persistent (Fig. 2C-D). Conversely, trees rarely stay in the high reproductive state for more than a single year, supporting the idea of non-mast years being more frequent than mast years (Kelly, 1994).

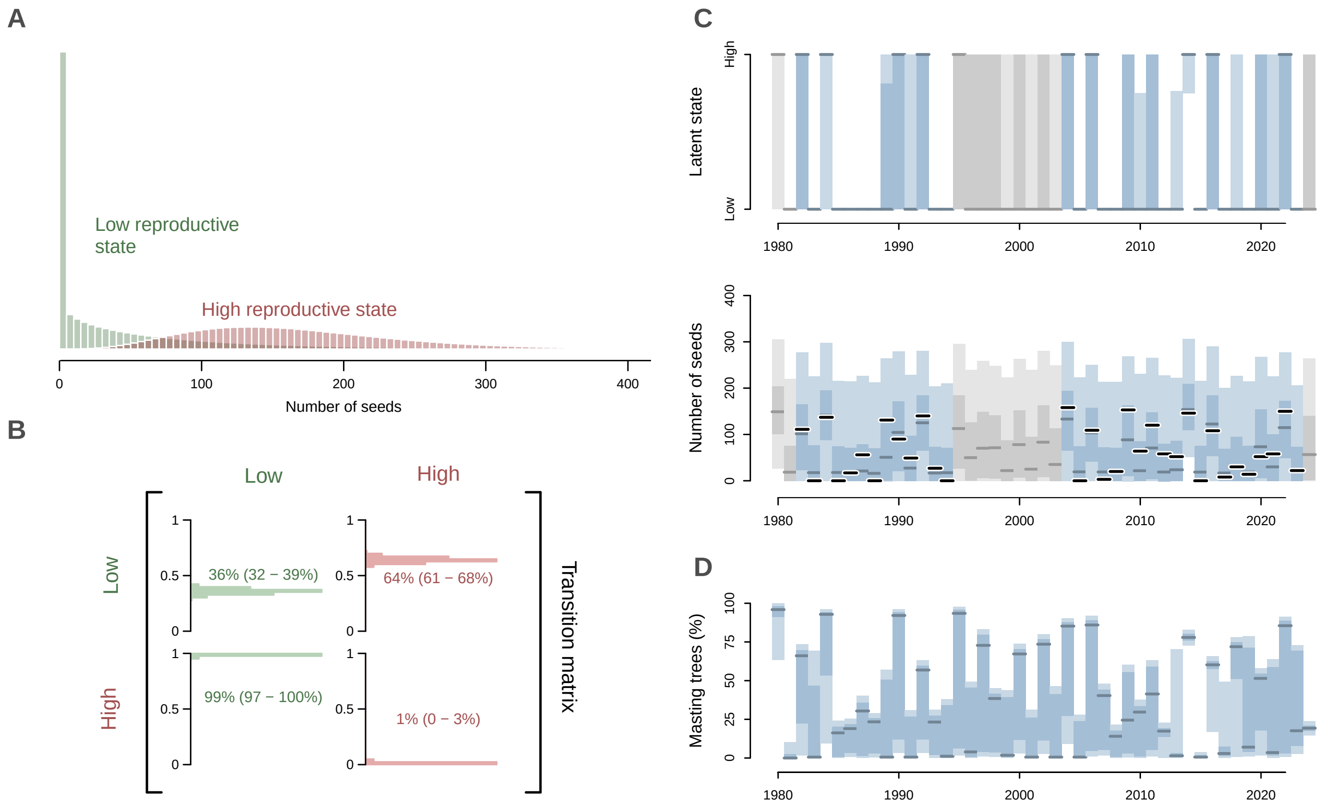

Figure 2: **Distinct tree reproductive states generates variability at the population level. (A)** The model identifies two distinct reproductive latent states that correspond to two different level of seed production. **(B)** The differences in transition probabilities between the two latent states reflect endogenous constraints that shape tree-level reproductive cycles. **(C)** At the tree level (illustrated here for one arbitrary tree), the alternation between these two latent states across time (upper plot) generates low and high seed production years (lower plot). **(D)** When added together across all populations, these individual cycles generate variability and synchrony in seed production across years. In **(C)** and **(D)**, the blue line represents the median model prediction, while the shaded areas represent the 50% and 90% uncertainty intervals. The black lines represent the observed seed counts, and the grey lines and areas indicate model predictions for years when observations were missing.

Climate strongly affected reproductive state transition dynamics. Similar to previous studies (Nussbaumer *et al.*, 2018; Vacchiano *et al.*, 2017; Journé *et al.*, 2024), we found that warm summer temperatures increased the probability that trees in a low reprodutive state transitioned to a high reproductive state (Fig. 3A). For example, the transition probability during a warm summer at 22°C was 3.6 times higher than during a 18°C summer (94% vs. 26% respectively, and ranges of 92 – 96% vs. 22 – 30%, Supp. Fig.). Additionally, our results support the common hypothesis that spring frosts may limit seed production (Augspurger, 2009; Packham *et al.*, 2012), with seed production for trees in a high reproductive state reduced by 20.1% (13.9 – 26.1%, Fig. 3B) given a three-fold increase in a metric of frost risk (growing-degree days until last frost, Vitasse & Rebetez, 2018), but no evidence that warm springs affected seed production (Supp. Fig.). Given the reproductive constraints that prevent beeches and many other woody plants from repetitive years of high fruit (from the competitive overlap of flower bud formation and fruit development, Monselise & Goldschmidt, 1982; Geber *et al.*, 1997), we assumed the transition from a high to a low reproductive state was not affected by climate (we found strong support for this when we relaxed this assumption, Supp. Fig.).

Individual tree-level reproductive states (high or low) scaled up to produce population-level seed production that was synchronous, both within and between populations (sites). Within populations, 87% (84 – 89%) of trees shared the same reproductive state on average each year, while synchrony across populations was only slightly lower (the proportion of trees in the same state across all populations was 84%, 81 – 86%). This high level of synchrony across populations could suggest that synchrony occurs across wide spatial scales, as sometimes suggested (Bogdziewicz *et al.*, 2021a; Journé *et al.*, 2024), but given that these populations were not widely separated spatially (Supp. Mat.), these results only provide evidence for synchrony over smaller spatial scales, as often suggested (Koenig & Knops, 1998; LaMontagne *et al.*, 2020). Whether this scale is sufficient for fitness benefits depends on the exact mechanism (e.g., predator satiation or pollination efficiency) and the species-level attributes (e.g. predators foraging ranges or wind dispersal distances, which depend on seed shape, size and the local landscape, Davies & MacPherson, 2024). For example, if beech masting in England mainly serves to overwhelm seed predators, such as wood mouse or bank vole (Sachser *et al.*, 2025), then the masting synchrony at the population level that we observe should be sufficient to maintain the regeneration benefits of masting (given that sites are on average 224 km apart, and the two closest sites 1 km apart).

Climatic cues—especially cold summers—appeared critical to synchronizing the reproductive cycle of individual trees to produce the population-level synchrony that often defines masting (Kelly & Sork, 2002). While 36% of trees in a low reproductive state are likely to stay in that state, cold summer temperatures prevent an additional subset from transitioning to a high reproductive state (because cold summer temperatures reduce the probability of transition to a high reproductive state, Fig. 3A). Thus natural reproductive constraints combined with cold summers can leave most trees in a low reproductive state. This synchrony of low reproduction appears to then drive synchrony of high reproduction when a cold summer is followed by a warm summer; in such years most trees responded consistently to the warm temperatures that promote floral induction, and transitioned together into a high reproductive state (Fig. 3C), thus generating a mast year at the population scale. This interplay between individual-level constraints and population-level climatic cues provides a physiological mechanism for reproductive synchrony that can persist over multiple years, and offers a simple explanation for previous findings that masting depends on the temperature difference between successive summers (the ΔT model; Kelly *et al.*, 2012).

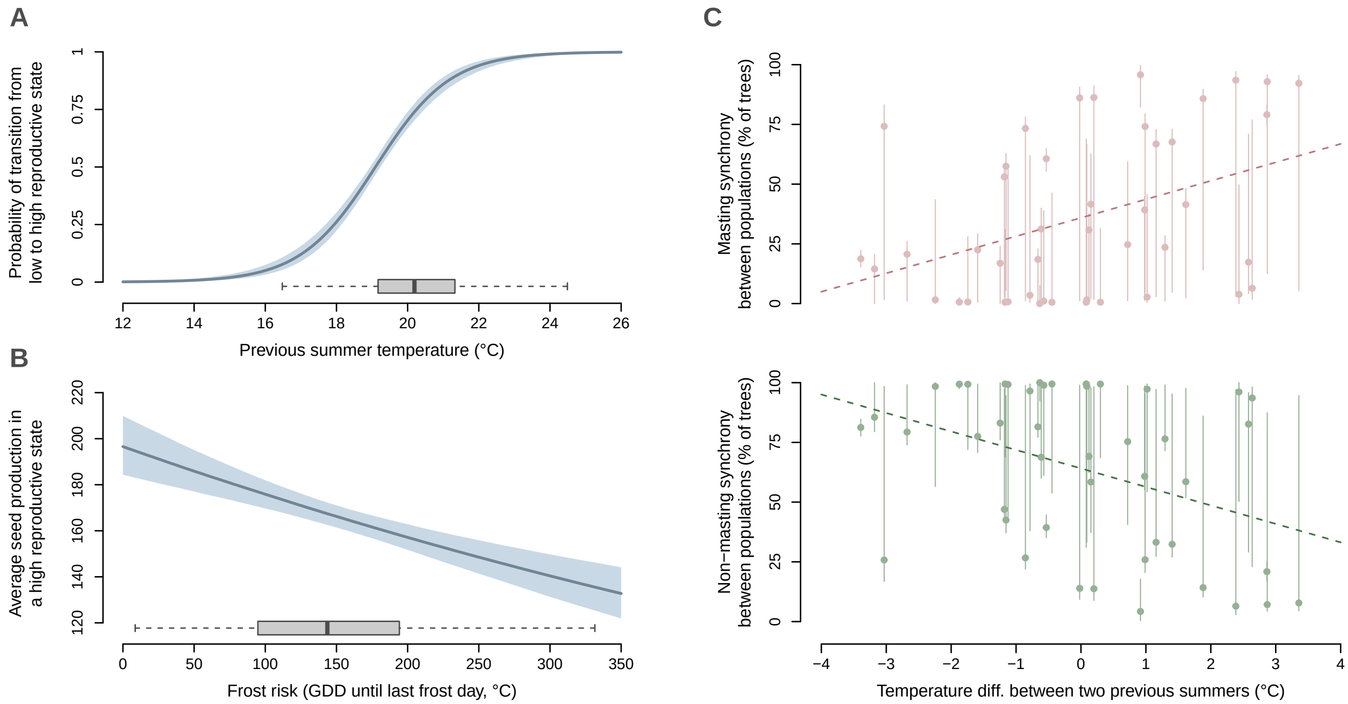

Figure 3: **Climate conditions impact both transition probabilities between reproductive states and seed production. (A)** Average maximum temperature in July and August of the previous year increases the probability of transition from a low to high reproductive state. **(B)** Frost risk (measured as accumulated growing-degree days, GDD, until the last frost day, see Vitasse & Rebetez, 2018).decreases the number of seeds produced when in a high reproductive state. **(C)** The interaction of previous-year summer conditions and latent reproductive states (top: high reproductive state, bottom: low reproductive state) explain the apparent two-year lag effect of climate on observed synchrony across populations. In **(A)** and **(B)**, the blue line and shaded area respectively represent the median and the 90% uncertainty interval, while the gray boxplots show observed conditions (1980–2022)

Our results appear to support the hypothesis that increasing summer temperatures with climate change could disrupt masting (Bogdziewicz *et al.*, 2021b; Foest *et al.*, 2024). This hypothesis—previously called breakdown—predicts that warm summers will drive trees into a high reproductive state more frequently and thus lead to fewer years of low reproduction. This would in turn reduce variability and synchrony in seed production at the population level, potentially minimizing the fitness benefits of masting, while also potentially leading to lower tree growth as trees invest in flowers and fruit most years (Koenig & Knops, 1998; Hacket-Pain *et al.*, 2025). While we do find that warmer summers increase the probability of trees transitioning into a high reproductive state—and thus increasing summer temperatures would increase the frequency of high seed production years—our results suggest that reproductive constraints prevent major shifts in the frequency of high seed production years. Indeed, we found even with extreme summer warming (7°C in one century, corresponding to the warmest regional climate model projections, Schumacher *et al.*, 2024), only 52% (40 – 64%) of trees in a population would mast in any given year (Fig. 4). This result remains even when we relax reproductive constraints and allow warm summers to increase the probability of trees staying a high-reproductive state (Supp. Fig.). Even with this additional effect, trees cannot become stuck indefinitely in a highly reproductive state. On the contrary, we found a 'breakdown' because of climate change would require summer warming to have a similar impact on both the transition and persistence into a high reproductive year—an hypothesis not supported by the data we used here (Fig. 4).

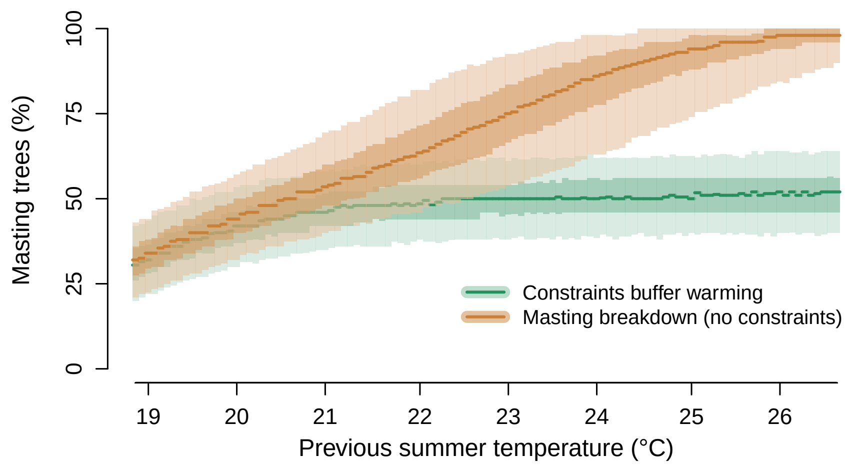

Figure 4: **Reproductive constraints prevent masting breakdown under a significant summer warming.** The green line and shaded areas (median, 50% and 90% uncertainty intervals) represent the predictions with the model fitted on the data used in this paper. With this model, the fitted transition probabilities indicate that trees cannot remain continuously in a high reproductive state. The orange line and shaded areas (median, 50% and 90% uncertainty intervals) represent the predictions from an hypothetical model where summer warming increases the probability of persisting into a high reproductive state—and would lead to most trees masting every year (i.e. reproductive breakdown). These predictions are not supported by the data used here.

These results highlight how biological constraints on plant reproduction may limit runaway effects of climate change. For masting, floral bud initiation at the individual tree level is promoted by warm summer temperatures, potentially leading to a higher number of flowers and fruits in the following year (Vacchiano *et al.*, 2017; Bogdziewicz *et al.*, 2024; Journé *et al.*, 2024). In the next summer, however, the development of a larger fruit load inherently imposes a trade-off that limits the tree from initiating a large number of floral buds again (Monselise & Goldschmidt, 1982; Wilkie *et al.*, 2008). These individual-level constraints scale up to the population level by preventing the entire population from producing large seed crops every year—and thus also preventing an unlimited amplification of climate effects on masting. Our results highlight that constraints at the individual level are critical for accurate forecasts of population-scale dynamics (Łomnicki, 1988). Further this approach can help identify the spatial scale at which changes occur, including possible shifts in synchrony.

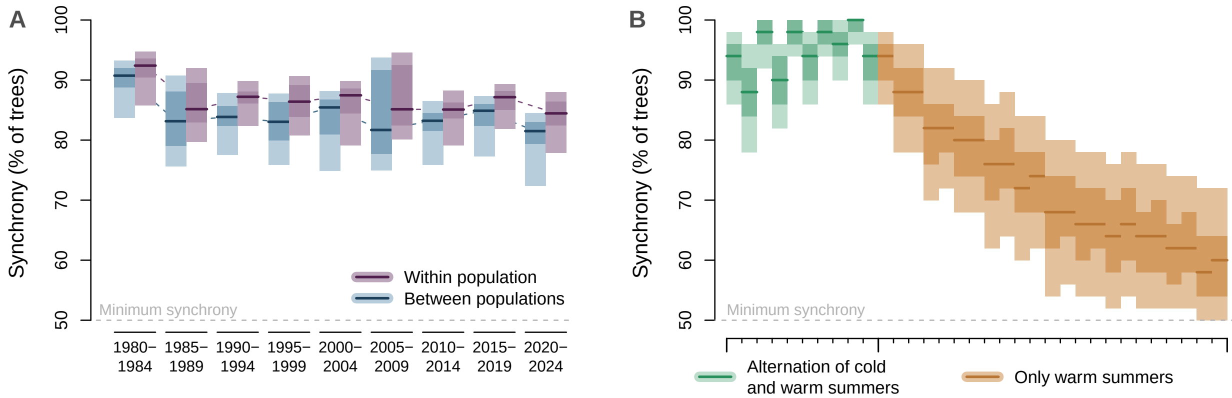

Figure 5: **Reproductive synchrony has remained largely stable over recent decades, but persistent decline in cold summers could desynchronize tree reproduction in the future. (A)** The lines (median) and shared areas (50% and 90% uncertainty intervals) represent the observed synchrony within population (in blue, estimated as the average proportion of trees of the same population that are in the same reproductive state) and across populations (in purple, estimated as the proportion of trees that are in the same state across all populations). **(B)** The lines (median) and shared areas (50% and 90% uncertainty intervals) represent the predicted percentage of masting trees in a hypothetical population over time (x-axis shows hypothetical years), where the first period includes alternating cold (around 17°C) and warm summers (around 22°C), in purple, and then experiences only warm summers, in green.

In contrast to previous results we found no clear evidence of declines in synchrony in recent decades, during a period of significant warming (Bogdziewicz *et al.*, 2020; Foest *et al.*, 2024). Synchrony across trees within a population varied over time but has not clearly changed in recent periods (recent declines in 2015 onward are within the 90% range of previous periods, Fig. 5A). Similarly, synchrony between populations has been quite stable, from a previous mean of 84.3% (81.1 – 87.2%) before 2015 to one of 82.8% (78.9 – 85.2%) since 2015. This stability in synchrony across both scales suggests that masting in these beech populations may be more resilient to warming than previously predicted (Bogdziewicz *et al.*, 2021b).

Whether masting can continue to support forest regeneration in the next decades, however, may depend on how synchrony responds as the climate continues to warm. While we find no evidence for a positive feedback of summer warming that would trigger runaway seed production at the individual level, our results predict possible changes at the population level in synchrony as we lose the cold summers. Because cold summers act as potential hinge points for population synchronization, our results suggest that several consecutive years without cold summers decrease synchrony within a population (Fig. 5B). Determining the extent to which this will happen in the coming decades, and producing robust spatiotemporal forecasts, will require better measurements of the temporal and spatial scales at which synchrony occurs (LaMontagne *et al.*, 2020). Our results show the value of building from individual-level seed production data, which is still incredibly rare, to understanding the population scale. Teasing out the full effects of warm summers versus spring frosts, however, will also require temporally finer resolution data than seed counts—including flowering (Nussbaumer *et al.*, 2020) and following from bud formation to flowering, thus including flower or fruit abortion (frost or drought damage of trees about to mast).

Fundamentally synchrony in seed production is one piece of a larger puzzle. Understanding future forest regeneration requires disentangling the full sequence of reproductive and growth phenology stages that lead from a floral bud to an established tree. New data on bud formation, flowering, and seed production across years could improve models of masting. Improving our understanding of this is the first critical step of a much longer sequence could help pinpoint how climate change will ultimately impact forest regeneration. Climate change may lead to major shifts in the later stages—seed germination, seedling establishment and tree recruitment (Petrie *et al.*, 2023), with potential global impacts on forests and the services they provide. Our results highlight that we may fail to accurately forecast these shifts until we integrate the biological and climatic constraints that start with individual trees.

Methods

Tree-level reproduction and climate data

We used annual seed production data from 171 individual beech trees (*Fagus sylvatica*) across 12 populations in England, collected between 1980 and 2024, following the protocol described in previous publications (Packham *et al.*, 2008; Bogdziewicz *et al.*, 2021b; Hacket-Pain *et al.*, 2025). Populations contained between 8 and 21 trees and were monitored for 5 to 45 years. Within populations, individual time series varied in length (average length = 20.4 years) and contained some missing observations.

We extracted daily climatic data from ERA5-Land (Muñoz-Sabater *et al.*, 2021), at a 0.01 degree resolution, for the 12 sites. We computed three climatic variables for each year of observation: average maximum temperature in July and August (of the previous year), average spring temperature (April-May), and growing degree days (GDD) until the last frost day as a measure of spring frost risk (Vitasse & Rebetez, 2018).

A Hidden Markov model for individual tree reproduction

Structure of the model

We modeled individual tree seed production with a Hidden Markov Model (HMM), in a Bayesian framework using the probabilistic programming language Stan (Carpenter *et al.*, 2017). In this model, each tree k is assigned to a latent reproductive state $z_{k,i}$ each year i: a low reproductive state ($z_{k,i} = 1$) or a high reproductive state ($z_{k,i} = 2$). The latent state $z_{k,i}$ is conditionally dependent only on the state the preceding year (the *Markovian* assumption), such that for N years we have:

$$p(z_{k,1},\dots,z_{k,N}) = p(z_{k,1}) \prod_{i=2}^{N} p(z_{k,i} \mid z_{k,i\text{-}1}) \tag{1}$$

The transitions between reproductive states are determined by a 2×2 matrix of probabilities:

$$\begin{bmatrix} p(z_{k,i}=1 \mid z_{k,i\text{-}1}=1) & p(z_{k,i}=2 \mid z_{k,i\text{-}1}=1) \\ p(z_{k,i}=1 \mid z_{k,i\text{-}1}=2) & p(z_{k,i}=2 \mid z_{k,i\text{-}1}=2) \end{bmatrix} \tag{2}$$

which is by definition equivalent to:

$$\begin{bmatrix} 1-p(z_{k,i}=2 \mid z_{k,i\text{-}1}=1) & p(z_{k,i}=2 \mid z_{k,i\text{-}1}=1) \\ 1-p(z_{k,i}=2 \mid z_{k,i\text{-}1}=2) & p(z_{k,i}=2 \mid z_{k,i\text{-}1}=2) \end{bmatrix} \tag{3}$$

Both the latent reproductive states and the transition probabilities were inferred from the seed data alone, i.e. the model identify the most probable state sequence for each individual tree given its observed seed counts.

The latent state determines the observational model responsible for the observed seed count $y_{k,i}$. We modeled the seed production in a low reproductive state with a zero-inflated negative binomial distribution, and the one in a high reproductive state with a negative binomial distribution, such as:

$$y_{k,i} \mid z_{k,i}=1 \sim \begin{cases} 0 & \text{with probability } \theta \\ \text{NegBin}(\lambda_{\text{low}}, \psi_{\text{low}}) & \text{with probability } 1-\theta \end{cases} \tag{4}$$

$$y_{k,i} \mid z_{k,i}=2 \sim \text{NegBin}(\lambda_{\text{high}}, \psi_{\text{high}}) \tag{5}$$

Influence of climate on state transition and seed production

We expected the climatic conditions experienced by a tree to influence both the transition matrix (Equation 3) and the seeds produced in a high reproductive state (Equation 5). Conversely, we assumed that climate does not impact seed production in a low reproductive state (Equation 4). In Equation 3, we assumed that only the transition from a low reproductive state to a high reproductive state was affected by climate (and thus, by construction, the probability of staying in a low reproductive as well):

$$p(z_{k,i}=2 \mid z_{k,i\text{-}1}=1) = \text{logit}^{-1}\left(\alpha_{1\to2} + \beta^{\text{summer}}_{1\to2} \cdot (X^{\text{summer}}_{k,i\text{-}1} - X^{\text{summer}}_{\text{baseline}})\right) \tag{6}$$

where $X^{\text{summer}}_{k,i\text{-}1}$ is the average maximum temperature during summer (the preceding year i-1), $\beta^{summer}_{1\to2}$ is the effect of these summer temperature, and $\text{logit}^{-1}(\alpha_{1\to2})$ is the transition probability when $X^{\text{summer}}_{t,i\text{-}1} = X^{\text{summer}}_{\text{baseline}}$.

We did not expect $X^{\text{summer}}_{k,i\text{-}1}$ to affect $p(z_{k,i}=2 \mid z_{k,i\text{-}1}=2)$ because the temporal overlap between fruit development and floral bud initiation should prevent trees already in a high reproductive state to respond to favorable summer conditions (though we still checked that this effect was negligible, see Supp. Fig.).

In Equation 5, we assumed that both spring temperature $X_{k,i}^{\text{spring}}$ and spring frost risk $X_{k,i}^{\text{frost}}$ could impact the seed production in a high reproductive state, such that:

$$y_{k,i} \mid z_{k,i} = 2 \sim \text{NegBin}(\lambda_{k,i}^{\text{high}}, \psi_{\text{high}}) \tag{7}$$

$$\log\left(\lambda_{k,i}^{\text{high}}\right) = \log(\lambda_{\text{high}}) + \beta_{\text{high}}^{\text{spring}} \cdot (X_{k,i}^{\text{spring}} - X_{\text{baseline}}^{\text{spring}}) + \beta_{\text{high}}^{\text{frost}} \cdot (X_{k,i}^{\text{frost}} - X_{\text{baseline}}^{\text{frost}}) \tag{8}$$

Masting synchrony and model predictions

Individual tree states were aggregated within and across populations to quantify reproductive synchrony, i.e. the proportion of trees sharing the same reproductive state (Fig. 5). Synchrony within populations was computed for each population as the maximum of the proportion of trees in a high or low reproductive state, then averaged across populations. Synchrony across populations was computed similarly, but the proportions of trees in each state were first averaged across all populations before taking the maximum (to avoid a population with more trees to have a higher contribution). Synchrony within populations can thus be higher than synchrony across populations when trees within a site tend to be in the same state but differ between sites. Since our model includes only two states, the minimum synchrony is 50% (i.e. at least 50% of the trees are in the same state).

To forecast the effects of summer warming (Fig. 4), we simulated seed production (and synchrony) for 50 trees representing an average English population with projected summer warming up to 7°C over one century (Schumacher *et al.*, 2024). To assess the role of biological constraints, we compared predictions from our model—in which reproductive constraints prevent trees to stay indefinitely in a high reproductive state—against a hypothetical model where summer temperatures had a similar effect on both $p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 1)$ (transition from low to high) and $p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 2)$ (persistence in high). To illustrate the role of cold summers in reproductive synchrony, we also simulated seed production for 50 trees first experiencing a perfect alternation of cold (around 17°C) and warm summers (around 22°C), followed by a period of warm summers only (Fig. **??**).

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